

When Yield Curves Invert Together: Predicting Currency Carry Losses

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Abstract

Currencies face country-specific and global risks. An inversion of one country's yield curve may reflect either, but synchronized inversions across several non-U.S. G10 economies are more likely to contain information about global conditions. I combine these inversions into a state variable that predicts unusually poor carry-trade spot returns. Annualized carry-trade spot returns are 12.9 percentage points lower when the lagged state is active than when inactive, with inference based on fifteen episodes. High-interest-rate currencies and currencies with greater predetermined equity-market betas experience larger losses, supporting the interpretation that the state contains information about global rather than country-specific risk.

JEL: E43, F31, G12, G15.

Keywords: currency carry trade, yield-curve inversion, exchange-rate predictability, global risk, G10 currencies.

1 Introduction

Currencies face both country-specific and global risks. This distinction is especially important for the carry trade. Diversification removes part of a currency’s idiosyncratic risk, but it does not protect a portfolio that is long several high-interest-rate currencies when those currencies depreciate together. Because interest income accrues gradually, one shared exchange-rate movement can erase many months of carry. The empirical question is whether bond markets reveal such a state before the currency loss occurs.

An inversion of one country’s yield curve is ambiguous. It may reflect domestic weakness, disinflation after a national tightening cycle, or a change in term premia. Synchronized inversions across several non-U.S. G10 economies are more likely to contain information about a global disturbance, although synchronization alone cannot distinguish global growth news from common policy or term-premium shocks. I combine fresh 10-year/2-year inversions across nine non-dollar G10 economies into a monthly state and ask whether that state predicts currency returns.

The construction is deliberately path dependent. A curve enters when it crosses into inversion and remains live until its slope has increased for two consecutive months. The synchronized-inversion state is active when at least two curves are live. Fresh entry separates new crossings from stale inversions; confirmed release keeps the state active when curves begin to re-steepen, including periods in which expected easing is being delivered. The state observed at the end of month t predicts the return during month $t + 1$.

The 1988–2026 sample contains 92 active months in fifteen episodes. Annualized carry-trade spot returns are 12.9 percentage points lower when the lagged state is active: 2.9 percent outside the state and -10.0 percent inside it. Total carry returns fall from 6.5 to -4.9 percent. The result is not only a label for the largest crashes. After removing each return series’ full-sample bottom-decile months, the spot-return gap remains -7.2 percentage points per year. The center of the distribution also shifts: the monthly carry-spot median falls from 0.35 to -0.76 percent.

The cross-section gives the result economic content. Carry buys high-interest-rate currencies and funds them with low-interest-rate currencies. Those rate rankings are correlated with exposure to global markets: the rank correlation between currencies’ average interest differentials and their average predetermined equity betas is 0.72. In active-state months, high-rate currencies depreciate relative to low-rate currencies,

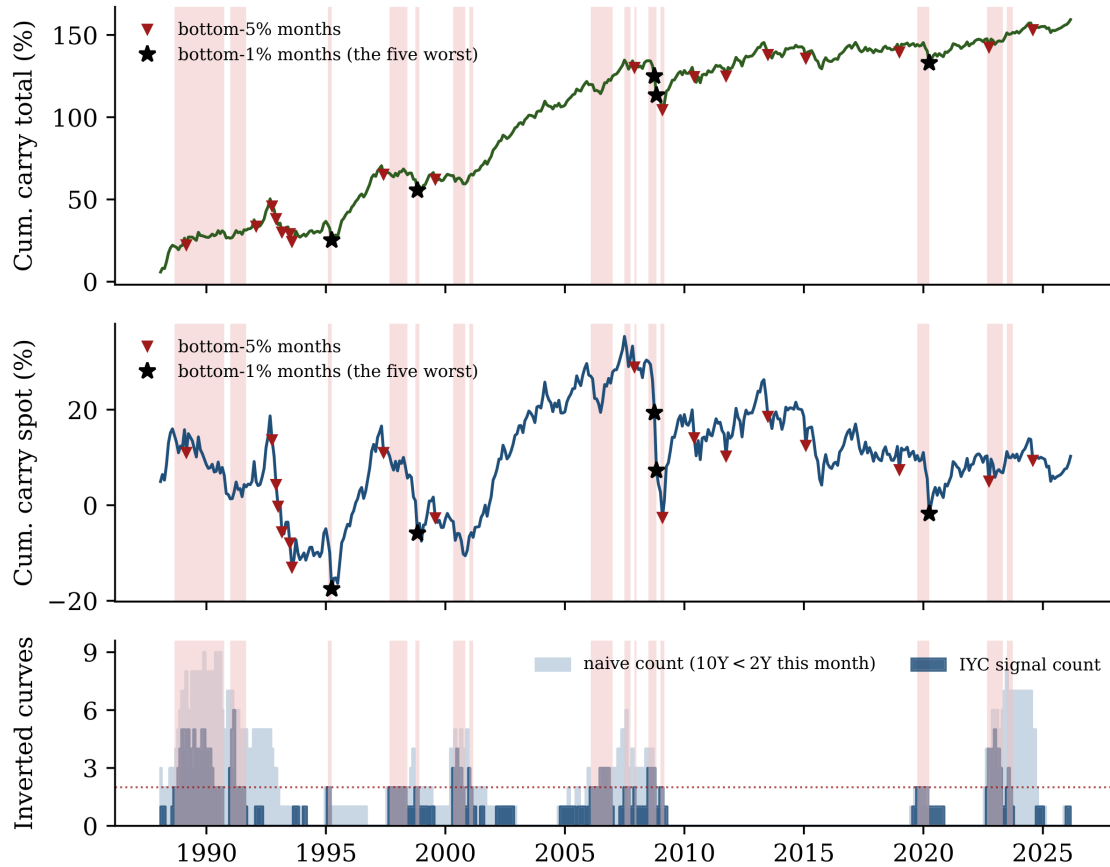


Figure 1: Carry returns and synchronized inversions. The light line counts curves currently inverted; the dark line counts fresh inversions that remain live under the release rule. The dotted line is the threshold of two. Shading denotes the end-of-month synchronized-inversion state, used with a one-month lag in every return regression. Tail markers show that the largest losses follow information already visible in the state, including episodes in which the current-inversion count has declined.

and bilateral losses become larger as predetermined equity beta rises. Across nine currencies, the beta gradient explains 77 percent of the variation in estimated state coefficients; a portfolio formed directly on expanding-window equity betas loses 7.5 percentage points per year in spot returns in the state. The panel interaction has the expected negative sign but weak episode-level inference. Taken together, the estimates support an exposure ordering, not a precisely identified structural beta.

This ordering links two established literatures. National yield slopes forecast activity and reflect both expected short-rate paths and term premia (Harvey, 1988; Estrella and Hardouvelis, 1991; Estrella and Mishkin, 1998; Campbell and Shiller, 1991; Hamilton and Kim, 2002). International term structures also contain a shared component (Plosser and Rouwenhorst, 1994; Bernard and Gerlach, 1998; Diebold et al., 2008; Chinn and Kucko, 2015). The carry literature, meanwhile, shows that high-interest-rate currencies earn average premia but suffer negatively skewed losses in common stress states (Fama, 1984; Brunnermeier et al., 2008; Lustig et al., 2011; Menkhoff et al., 2012; Farhi and Gabaix, 2016). The contribution here is to shift the unit of observation from one country’s slope or a smooth global level factor to a cross-country configuration of fresh threshold crossings dated before the currency payoff.

The exposure ordering also clarifies why interest rates matter twice. The differential is the carry trade’s known income, but it also ranks the currencies that bear the largest state-contingent spot losses. Commodity exposure, country size, trade-network position, and intermediary demand can all align high rates with sensitivity to global conditions (Hassan, 2013; Ready et al., 2017; Richmond, 2019; Gabaix and Maggiori, 2015). The paper does not distinguish those channels. It shows that the loss is concentrated in currencies for which a predetermined market measure indicates greater exposure, rather than appearing as a uniform dollar movement.

Several results delimit this interpretation. One live inversion has the opposite return sign from two; the U.S. inversion alone has a small and imprecise coefficient; and the continuous average G10 slope, lagged financial conditions, and lagged VIX do not span the synchronized state. The five worst monthly carry returns occur under the lagged state, yet trimmed estimates and calendar splits show that 2008 and 2020 are not the entire result. These comparisons distinguish synchronization from nearby curve and risk measures. They do not identify the shock that synchronizes curves, and they do not remove the central design risk: the tenor, threshold, and release rule were assessed on the same historical sample.

Finally, a predictable negative conditional return is not an automatic implication of disaster exposure. In a frictionless benchmark, publicly known bad-state exposure should command compensation. The paper therefore treats slowly adjusting compensation as one candidate interpretation: fresh bond-market information can raise perceived loss risk before currency compensation fully adjusts. This mechanism explains why entry and timing may matter, but the return evidence does not identify the friction. The durable contribution is narrower and more testable: synchronized foreign yield-curve inversions predict where common currency losses subsequently occur.

Section 2 defines returns, the state, and timing. Sections 3–4 present the design and evidence. Section 5 develops the exposure ordering and economic interpretation; Section 6 states the main limits. The online appendix gives the exact algorithm, derivations, supplementary estimates, and public-data checks.

2 Measurement and Timing

2.1 Currency returns and the shared-loss problem

Consider a U.S. investor holding currency i . Let $s_{i,t}$ be the log dollar price of one unit of foreign currency, so $\Delta s_{i,t+1} > 0$ is foreign-currency appreciation. Let $d_{i,t} \equiv (i_{i,t} - i_{US,t})/12$ be the one-month interest differential, formed from annualized money-market rates. The excess-return proxy is

$$rx_{i,t+1} = d_{i,t} + \Delta s_{i,t+1}. \quad (1)$$

The differential is known at t ; the spot change is realized from t to $t + 1$. Under covered interest parity this sum corresponds to a forward return apart from basis and measurement differences (Du et al., 2018). Because the paper uses money-market differentials rather than executable forward quotes, the object is an asset-pricing return proxy, not a transaction-cost-adjusted strategy.

Each month, the carry portfolio buys the three highest-rate currencies and sells the three lowest-rate currencies among the nine non-dollar G10 currencies. If $\mathcal{H}_t$ and $\mathcal{F}_t$ denote the high-rate and funding sets, then

$$\begin{aligned} rx_{t+1}^{\text{carry}} &= \frac{1}{3} \sum_{i \in \mathcal{H}_t} rx_{i,t+1} - \frac{1}{3} \sum_{i \in \mathcal{F}_t} rx_{i,t+1} \\ &\equiv C_t + X_{t+1}. \end{aligned} \quad (2)$$

Here C_t is the long-minus-short interest differential and X_{t+1} is the corresponding spot return. The sample-average income leg is 3.9 percent per year and is positive in every month, with a minimum of 0.7 percent. Carry therefore loses only when the spot leg is more negative than the income leg is positive.

To organize the cross-sectional test, write the schematic decomposition

$$\Delta s_{i,t+1} = -\lambda_i g_{t+1} + e_{i,t+1}, \quad (3)$$

where g_{t+1} is an adverse return component shared across currencies, λ_i is currency i 's latent exposure, and $e_{i,t+1}$ is idiosyncratic. Diversification reduces the idiosyncratic component but not the shared component when the two portfolio legs have different average exposures. Here “shared” is statistical; “global” is the economic interpretation.

Equation (3) is an organizing device, not an estimated structural model.

2.2 The synchronized-inversion state

Let $q_{i,t} \equiv y_{i,t}^{10Y} - y_{i,t}^{2Y}$ denote country i 's end-of-month yield slope. An inversion can reflect expected easing, term-premium variation, or both (Campbell and Shiller, 1991; Hamilton and Kim, 2002; Bauer and Mertens, 2018). Several fresh crossings are more suggestive of a shared disturbance than one crossing, although regional shocks and coordinated policy can also synchronize curves.

Define $I_{i,t} = \mathbf{1}\{q_{i,t} < 0\}$. A curve enters the live set when it freshly crosses from $I_{i,t-1} = 0$ to $I_{i,t} = 1$. It remains live until $q_{i,t}$ has increased for two consecutive months. A released curve cannot re-enter while still inverted; it must first become nonnegative and then cross below zero again. Let $L_{i,t}$ be the resulting live status. The central state is

$$N_t = \sum_{i=1}^9 L_{i,t}, \quad S_t = \mathbf{1}\{N_t \geq 2\}. \quad (4)$$

Thus $L_{i,t}$ is path dependent, whereas $I_{i,t}$ depends only on the current slope. The state measured at the end of month t conditions the return during month $t + 1$.

The live rule has two economic roles. Fresh entry excludes inversions already embedded in prices for many months. Confirmed release allows the state to remain active as curves re-steepen, which can coincide with delivered easing and delayed currency adjustment. In the sample, a raw two-or-more current-inversion state is active in 165 months, compared with 92 for the live state, yet switches off before March 1995, October 2008, and March 2020—the three largest carry-loss months. This historical fit is informative and also creates specification-search risk. Online Appendix A states the transition rules and boundary cases exactly.

2.3 Data, information set, and predetermined exposure

The monthly sample runs from January 1988 through February 2026. The currency universe comprises the Australian dollar, Canadian dollar, Swiss franc, euro (Deutsche mark before 1999), British pound, Japanese yen, Norwegian krone, New Zealand dollar, and Swedish krona. The dollar is the numeraire and does not enter either the sort or the baseline state count. The U.S. curve is retained as a benchmark.

State incidence varies across monetary regimes. It is active in 33 of 84 months during

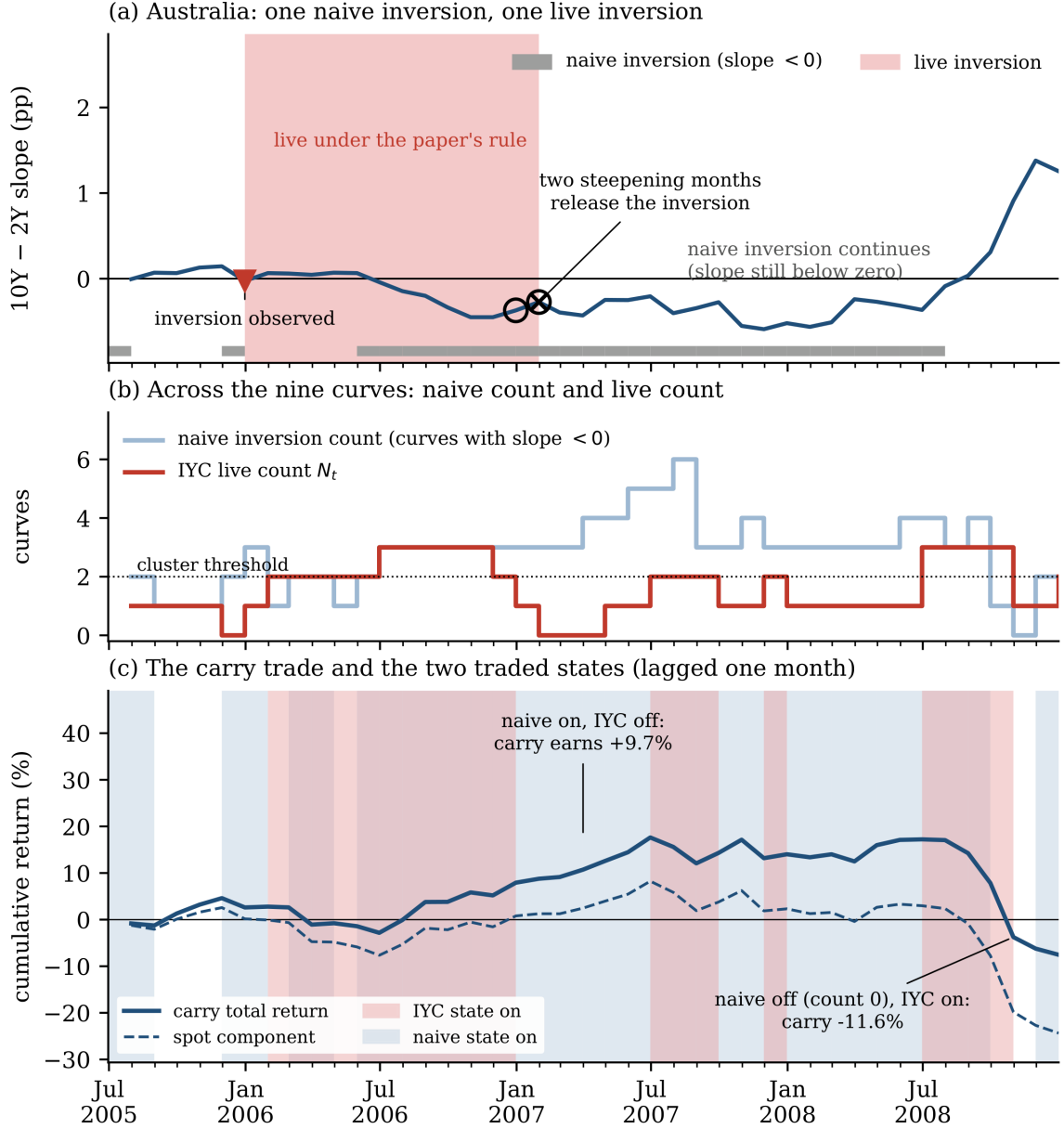


Figure 2: Live-state construction, 2005:07–2008:12. Panel (a) shows the Australian 10Y–2Y slope and its live status. Panel (b) compares the current-inversion and live counts. Panel (c) plots cumulative carry returns under the two lagged states. Fresh entry removes stale inversions; confirmed release can retain the state after the current-inversion count declines.

1988–94 but only 3 of 120 months during 2010–19, when many policy rates were near their lower bounds. The state is therefore not a continuously available global factor; it appears mainly when international rate cycles create enough cross-country slope variation. Online Appendix A reports the full era counts.

The baseline state uses end-of-month 2-year and 10-year government yields. Portfolio weights use one-month money-market rates known at t . Return regressions use no information dated after t to predict month $t + 1$. This convention prevents mechanical look-ahead but does not eliminate asynchronous market closes, stale quotes, or later data revisions.

The exposure test estimates a currency’s equity beta on an expanding window:

$$rx_{i,\tau} = a_i + \hat{\beta}_{i,t} r_\tau^M + u_{i,\tau}, \quad \tau \leq t - 1. \quad (5)$$

The estimate $\hat{\beta}_{i,t}$ is available before the return from t to $t + 1$ and requires at least 36 prior monthly observations. The one-month buffer follows the original construction: the estimate paired with S_t uses currency and equity returns only through $t - 1$. The beta is an empirical proxy for the latent exposure λ_i in equation (3); it need not equal that primitive. S&P 500 total returns form the market measure.

Table 1: Notation and timing

Symbol	Definition	Timing
$t; t + 1$	Information month-end; subsequent holding month	Information $\rightarrow$ return
$s_{i,t}; \Delta s_{i,t+1}$	Log dollar price of currency i ; its one-month change	Spot change realized in $t + 1$
$d_{i,t}$	Monthly foreign-minus-U.S. interest differential	Known at t
$rx_{i,t+1}$	Currency excess-return proxy	Realized in $t + 1$
$C_t; X_{t+1}$	Carry income leg; carry spot leg	Known at t ; realized in $t + 1$
$q_{i,t}$	Country i ’s 10Y–2Y yield slope	Observed at month-end t
$I_{i,t}; L_{i,t}$	Current inversion; path-dependent live inversion	Information at t
$N_t; S_t$	Live-curve count; synchronized-inversion state	Predict $t + 1$
$\lambda_i; \hat{\beta}_{i,t}$	Latent shared-factor exposure; estimated equity-beta proxy	Beta uses data through $t - 1$
W_t	Vector of lagged benchmarks or controls	Known at t

Notes: Equations use the information-time convention $t \rightarrow t + 1$. Tables indexed by the return month equivalently report S_{t-1} .

3 Empirical Design

3.1 Predictive regression and inference

All return tests use one timing convention: information at t predicts the return at $t + 1$. The general specification is

$$Y_{t+1} = \alpha + bS_t + \Gamma'W_t + \varepsilon_{t+1}, \quad (6)$$

where W_t contains the relevant lagged benchmarks or controls. The headline conditional-mean difference sets W_t empty. With a binary state, b is the active-minus-inactive mean difference; monthly return coefficients are multiplied by twelve for reporting. In tables indexed by return month, equation (6) is equivalently written with S_{t-1} .

The state’s 92 active months form fifteen contiguous episodes, fourteen longer than one month. Treating all active months as independent would overstate the information in the sample. I report Newey–West statistics with 12 lags and, for binary-state tests, a wild episode bootstrap that assigns Rademacher weights to contiguous episodes and imposes the null. The episode bootstrap is the headline inference because continuation months do not create independent synchronized shocks. Fifteen episodes nevertheless provide limited common-state variation; no resampling method can manufacture a missing holdout.

Distributional diagnostics report medians, lower quantiles, skewness, expected shortfall, and conditional variance. A separate diagnostic removes each outcome’s full-sample bottom-decile observations from both states before re-estimating equation (6). Because that deletion conditions on realized returns, it is not a preferred estimator. It answers only whether a handful of extreme observations mechanically generate the mean gap.

3.2 Benchmarks and controls

The benchmark hierarchy moves from nearby measurement alternatives to broader risk states. The raw current-inversion state asks whether fresh entry and confirmed release add information. The U.S. inversion asks whether the conventional single-country proxy spans the international state. The average G10 slope and its change ask whether the threshold is merely a nonlinear transformation of a smooth global slope. Nine country-specific live states ask whether synchronized crossings add information beyond

their individual components.

Lagged NFCI and VIX controls address predictive spanning: do variables already known at t absorb the state’s forecast? Contemporaneous dollar and currency-volatility factors answer a different question: do those realized factors account for the loss after it occurs? Neither design identifies the shock represented by S_t . Joint specifications are reported without sequential orthogonalization because the ordering of closely related yield-curve transformations would otherwise determine which variable receives their shared variation.

3.3 Exposure ordering

The bilateral state coefficients come from

$$\Delta s_{i,t+1} = a_i + b_i S_t + e_{i,t+1}. \quad (7)$$

The nine estimates $\hat{b}_i$ are compared with currencies’ average predetermined equity betas. The slope, fitted intercept at zero beta, and uncertainty describe the exposure gradient. A beta-sorted portfolio and a panel interaction $S_t \times \hat{\beta}_{i,t}$ provide complementary tests: the portfolio is economically transparent; the panel uses monthly variation but still has only fifteen common-state episodes.

This design separates three claims. Prediction asks whether S_t forecasts returns. Exposure accounting asks whether the realized losses are ordered by a predetermined market characteristic. Mechanism identification would require independent variation that distinguishes global growth, policy, term-premium, inflation, and intermediary channels. The first two are directly estimated; the third is not.

3.4 Failure patterns

Four findings would materially weaken the paper: the return gap disappearing after tail removal; one-country or smooth-slope benchmarks spanning the state; no relation between currency losses and predetermined exposure; or a real-time implementation requiring future observations. The reported estimates run counter to these patterns, but all variants were examined on the same history. A prospective frozen-rule evaluation remains the decisive external test.

4 Main Evidence

4.1 A large spot-return shift

Annualized carry-trade spot returns are 12.85 percentage points lower when the lagged synchronized-inversion state is active, with inference based on fifteen episodes. Spot returns average 2.85 percent in the 366 inactive months and -10.00 percent in the 92 active months. Total carry returns fall from 6.45 to -4.86 percent, a difference of -11.31 percentage points. Episode-bootstrap p -values are 0.001 and 0.003.

Table 2: The carry trade in and out of the IYC signal

Portfolio component	Annualized mean		State regression	
	No signal	Signal	Difference	Episode wild p
Carry spot	2.85	-10.00	-12.85	0.001
Carry total	6.45	-4.86	-11.31	0.003
High-beta spot	1.26	-6.27	-7.53	0.010
High-beta total	3.18	-3.64	-6.82	0.013

Notes: Returns are annualized percentages. The no-signal state contains 366 months and the signal state 92 months over 1988:01–2026:02. The difference is the coefficient on the one-month-lagged IYC-state indicator. Wild-bootstrap p -values cluster by the state’s fifteen contiguous episodes ($B = 9,999$).

The accounting matters. Carry income averages 3.9 percent per year and remains positive in every month, yet it cushions only part of the active-state depreciation. The result is therefore not low carry income; it is a spot movement large enough to overwhelm known income. The strategy’s annualized Sharpe ratio falls from 0.89 outside the state to -0.51 inside it.

The distribution shifts beyond the mean. The monthly spot median falls from 0.35 percent outside the state to -0.76 percent inside; the 25th percentile moves from -0.93 to -1.98 percent, and the 10th percentile from -2.35 to -3.94 percent. Spot-return expected shortfall worsens from -4.80 to -8.14 percent. By contrast, the conditional variance ratio is 1.74 with permutation $p = 0.14$. Location and tail frequency shift, while the increase in variance is not statistically detectable.

Removing each return series’ 46 bottom-decile months leaves a carry-spot coefficient of -7.24 percentage points per year ($p = 0.006$) and a total-return coefficient of -5.50 ($p = 0.024$). This outcome-selected diagnostic does not estimate a counterfactual without crashes. It shows that the mean difference is not arithmetically carried by only

the most extreme observations.

4.2 Why synchronization matters

The count displays a striking sign change. In 88 months with exactly one live inversion, carry spot and total returns are 8.33 and 8.78 percentage points per year higher than in the remainder of the sample, with episode-bootstrap p -values of 0.005 and 0.003. At exactly two live inversions, the coefficients reverse to -15.72 and -15.13 percentage points, each with $p \leq 0.003$. This contrast supports the economic distinction between one-country and synchronized information within the sample. It does not identify two as a universal cutoff.

Nearby bond-market states do not reproduce the result. The U.S. 10-year/2-year inversion is active in 57 lagged months and has carry-spot and total-return coefficients of -1.90 ($p = 0.484$) and -1.34 ($p = 0.637$). Adding the average G10 slope, its one-month change, and the U.S. inversion leaves the synchronized-state coefficient at -13.45 for spot returns ($p = 0.003$) and -12.44 for total returns ($p = 0.012$). The continuous slope variables are imprecise.

Adding all nine country-level live states makes the synchronized coefficient more negative. That increase is not evidence of a cleaner causal effect: the regressors are related transformations of a small set of curves, and individual states absorb many benign one-country episodes. The defensible conclusion is that neither a smooth global slope nor the individual curve states span the cross-country configuration.

4.3 Risk controls, timing, and stability

Lagged observed risk conditions do not absorb the state. With the NFCI level and change, the carry-spot coefficient is -12.84 percentage points ($p < 0.001$). Adding VIX level and change in the post-1990 sample gives -13.18 ($p = 0.004$). Contemporaneous dollar and global-FX-volatility factors reduce the coefficient only to -11.5 ($p = 0.002$). The lagged controls address forecast spanning; the contemporaneous controls show that the portfolio loss is not merely the average dollar movement or the particular volatility innovation in those regressions.

A U.S. term-premium adjustment gives a more mixed result than the simple controls. After stripping each national slope's fitted exposure to the U.S. ACM slope premium, 83 percent of inversions remain. The high-beta state effect remains negative and precise

(-6.2 percentage points, $p = 0.004$), whereas the carry effect roughly halves to -5.4 and is imprecise ($p = 0.11$). Thus a common U.S. term-premium component does not explain the exposure result, but it may account for part of the carry estimate; it is not a decomposition of foreign-country term premia.

The chronology is genuinely predictive under the stated month-end convention. All five worst monthly carry returns occur under the lagged state. March 1995, October 2008, and March 2020 arrive after the raw current-inversion count has receded but while the previously observed live state still conditions the return. Applying the rule recursively produces no later revisions to prior classifications. One additional month of lag roughly halves the carry coefficient, consistent with short-lived information rather than a permanent regime.

Calendar and episode checks are supportive but not external validation. The carry-spot coefficient is -11.41 percentage points before 2005 and -15.85 afterward. Leave-one-episode-out carry estimates remain within roughly one-third of the full-sample estimate. The beta-sorted portfolio is less stable across halves, with its adverse association concentrated after 2005. Online Appendix C reports these results and the rule grids in full.

The main evidence therefore supports an economically large within-sample predictive association: synchronized fresh inversions forecast lower subsequent carry spot returns, and the result is not reproduced by the most obvious single-curve, smooth-slope, or lagged-risk alternatives. The next section asks why the losses fall disproportionately on the high-rate currencies that carry holds.

5 Where the Losses Land

5.1 Rates and predetermined equity exposure

The carry result already contains one cross-sectional ordering: the portfolio loses because the high-rate currencies it buys depreciate relative to the low-rate funding currencies. The bilateral estimates in equation (7) provide a second ordering. State coefficients become more negative as currencies' predetermined equity-market betas rise.

Australia and New Zealand have average betas of 0.20 and 0.13 and annualized state spot coefficients of -13.21 and -13.28 percentage points. Japan and Switzerland have

betas near zero and imprecise coefficients of 2.35 and -0.89 . Across all nine currencies, regressing the state coefficient on average beta gives a slope near -60 percentage points per beta unit, an intercept of -0.75 , and $R^2 = 0.77$. With nine observations and no displayed confidence region for the fitted intercept, this is descriptive evidence of a gradient; it does not show that all global exposure vanishes at zero equity beta.

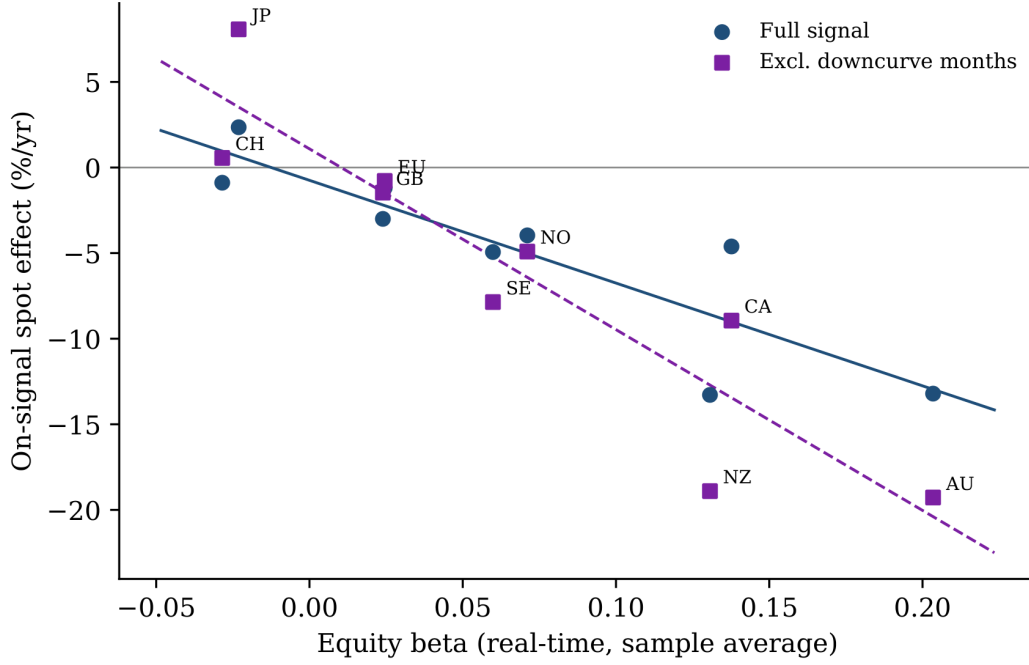


Figure 3: State-dependent spot-return coefficients and predetermined equity beta, one point per currency. Circles and the solid fit show the baseline state (slope -60 , $R^2 = 0.77$). Squares and the dashed fit exclude exploratory downcurve months. The baseline inference uses the solid fit.

Table 3: How active-state losses align with exposure

Object	Estimate	Episode wild p	Interpretation
Rate–beta rank correlation	0.72	—	Descriptive, nine currencies
Cross-currency beta slope	−60	—	Percentage points per beta unit
Cross-currency fitted intercept	−0.75	—	Percentage points at zero beta
Cross-currency fit	$R^2 = 0.77$	—	Descriptive, nine currencies
Beta-sorted spot portfolio	−7.53	0.010	Annualized state difference
Panel state $\times$ beta	−27	0.210	Annualized interaction
Conditional Fama slope, active state	−1.33	0.340	p -value is for the state interaction

Notes: The cross-currency slope, intercept, and R^2 come from regressing the nine bilateral state coefficients on currencies’ sample-average predetermined equity betas. The beta-sorted portfolio uses expanding-window betas. The panel interaction uses monthly predetermined betas. The Fama row is supporting conditional-moment evidence; its episode-level uncertainty is wide.

The beta-sorted portfolio uses a different predetermined ranking variable but the same underlying currency histories. Its spot return is 7.53 percentage points per year lower in the state, with episode-bootstrap $p = 0.010$. The panel interaction between the state and monthly predetermined beta is also negative, about −27 percentage points per beta unit, but has $p = 0.210$. Thus the carry portfolio establishes the return fact, the beta portfolio supports exposure ordering, the nine-currency scatter is descriptive, and the panel interaction is imprecise.

The high-beta distribution also differs from carry’s broad location shift. Its probability of a bottom-decile month rises from 8.5 percent outside the state to 16.3 percent inside ($p = 0.046$), while its variance ratio is 1.05 ($p = 0.80$). The active state is associated with a higher frequency of adverse returns, not a statistically detectable increase in unconditional variance. The cutoff and several distributional tests were examined jointly, so this remains supporting evidence rather than a separate discovery.

The ordering reflects persistent portfolio membership rather than a wholesale signal-month rotation. New Zealand occupies the carry long leg in 85 percent of all months and 91 percent of active-state months; Australia does so in 71 and 73 percent. Japan and Switzerland remain the core funding currencies. Thus the state primarily reprices standing cross-country exposure rather than changing which currencies the sort holds.

The economics is consistent with several established explanations for persistent cur-

rency rankings. Small-country exposure, commodity-export sensitivity, trade-network position, and global intermediary demand can all make high-rate currencies covary more positively with global conditions (Hassan, 2013; Ready et al., 2017; Della Corte et al., 2016; Richmond, 2019; Hassan and Mano, 2019; Gabaix and Maggiori, 2015). Common currency factors and business-cycle exposure provide related asset-pricing interpretations (Lustig and Verdelhan, 2007; Lustig et al., 2011; Lettau et al., 2014; Colacito et al., 2020). The rate-beta rank correlation is 0.72 in this sample; it does not establish that one primitive channel generates both rankings.

5.2 Spot repricing versus predetermined compensation

Seven floating emerging-market currencies provide a directional out-of-construction check because none enters the G10 state definition. All seven bilateral state coefficients are negative. Pooling them gives annualized spot and total-return effects of -14.37 and -13.35 percentage points, with episode-bootstrap p -values of 0.104 and 0.079. The signs and magnitudes are consistent with a broad international state, but the sample overlaps the G10 episodes and is neither an independent signal replication nor a frozen holdout (Bansal and Dahlquist, 2000; Ilzetzki et al., 2019).

The combined G10-plus-emerging-market carry portfolio isolates the role of compensation. On the common 1996–2026 sample, active-state spot returns are almost identical for G10 carry and G10-plus-emerging-market carry: -12.79 and -12.51 percent per year. Total returns differ: -8.67 for G10 and $+2.51$ for G10 plus emerging markets. The implied active-state income legs are approximately 4.12 and 15.02 percent. The state still lowers the broader portfolio’s total return by 9.18 percentage points relative to inactive months, but that estimate is imprecise ($p = 0.235$).

This comparison separates two economic objects that the baseline G10 portfolio alone cannot. Synchronized curves locate a similar international spot repricing across the two carry universes; predetermined interest income determines whether that spot loss produces a negative total mean. The state is therefore more naturally interpreted as a spot-risk state than as a universal negative-total-return rule. Online Appendix C reports the common-sample comparison.

5.3 Supporting conditional moments

Two additional results are economically coherent but statistically weaker. In a conditional Fama regression, the pooled appreciation slope on the lagged interest differential moves from 0.50 outside the state to -1.33 inside it. High-rate currencies therefore depreciate in the active state in point estimates, but the episode-bootstrap p -value for the interaction is 0.340. The result is a descriptive regression image of the carry loss, not precise evidence that uncovered interest parity is restored.

The OECD composite leading indicator declines about 0.6 percent over six months after a state month and remains similar around episode onsets and with lag controls. Industrial production also declines in baseline projections but becomes imprecise with the fullest controls. The leading-indicator response is consistent with the yield-curve literature’s expected-slowdown interpretation (Harvey, 1988; Estrella and Hardouvelis, 1991; Estrella and Mishkin, 1998); the estimates do not establish that synchronized inversions cause activity to fall. Online Appendix C reports the conditional-UIP and local-projection details.

5.4 A compact compensation benchmark

A predictable negative conditional return is not implied mechanically by bad-state exposure. The frictionless Euler condition $\mathbb{E}_t[m_{t+1}rx_{t+1}] = 0$ says that exposure to high-marginal-utility states should command compensation. To state the timing problem compactly, let an adverse event of size $J > 0$ occur at $t + 1$ with conditional probability p_t . Suppose the known carry income can be written as $C_t = \bar{C}_t + \mu J \mathbb{E}_{t-1}[p_t]$, where $\bar{C}_t$ is ordinary carry income and the second term is compensation based on the earlier information set, with $\mu \geq 1$. Then

$$\mathbb{E}_t[rx_{t+1}] = \bar{C}_t + J(\mu \mathbb{E}_{t-1}[p_t] - p_t). \quad (8)$$

A negative conditional mean requires the current perceived loss probability to exceed both previously priced risk and ordinary carry income: $p_t > \mu \mathbb{E}_{t-1}[p_t] + \bar{C}_t/J$. Fresh synchronized inversions can produce that inequality if bond-market information changes before currency compensation fully adjusts. Slow portfolio adjustment or constrained intermediary capacity are possible frictions (Bacchetta and van Wincoop, 2010; Gabaix and Maggiori, 2015); the reduced-form evidence does not identify either one. Online

Appendix B gives the payoff derivation.

5.5 Alternative common-state interpretations

Expected-rate news is one candidate. Synchronized curves can reflect coordinated changes in expected policy paths, but lagging the state establishes chronology rather than exogeneity. Common inflation news is another: international yield-curve components contain substantial inflation variation, and disinflation can synchronize inversions without a global growth shock (Jotikasthira et al., 2015). A common term-premium shock is also possible because an inversion is not a pure expectations measure. The asymmetric ACM result—a robust high-beta effect but an attenuated carry effect after the adjustment—keeps that channel open.

Intermediary constraints can jointly affect bonds and exchange rates. Lagged NFCI and VIX do not span the state, but those variables are incomplete proxies for dealer balance sheets, funding conditions, and global risk-bearing capacity (Brunnermeier et al., 2008; He et al., 2017; Miranda-Agrippino and Rey, 2020). Finally, heterogeneous responses to a broad dollar shock can generate some beta ordering even when the average dollar factor is controlled. The evidence makes a uniform one-for-one dollar account less plausible; it does not eliminate every dollar-based mechanism.

The economic conclusion is comparative. The state predicts losses that are ordered by both interest rates and predetermined equity exposure, and the fitted response is small for near-zero-beta currencies. That pattern supports the interpretation that synchronized inversions contain information about a shared international risk rather than only separate national disturbances. It does not identify the primitive shock or the friction that prices it.

6 Robustness and Limits

The robustness evidence answers three questions. The median, lower quantiles, and bottom-decile deletion show that the return gap extends beyond a few crisis observations. Raw inversion breadth, the U.S. curve, the continuous G10 slope, NFCI, and VIX do not reproduce the result in the reported specifications. Carry coefficients remain negative in both calendar halves and within roughly one-third of the full-sample estimate when any one episode is removed.

An exploratory public-data screen of all two- and three-country delivered-easing combinations produces one family-adjusted CHF–GBP lead, but it weakens under stricter event-count rules and after omitting October 2008. Online Appendix E therefore treats it as a sensor-versus-risk-bearer hypothesis, not a country-driver result.

The exact-one versus exact-two contrast isolates synchronization within the sample. The high-beta portfolio and country gradient support exposure ordering, although the panel interaction is imprecise. Replacing 10Y–2Y slopes with 10Y–3M slopes or removing a U.S. term-premium component attenuates the carry estimate but preserves a statistically detectable high-beta loss. Exposure ordering is less sensitive to these perturbations than the baseline carry magnitude. Conditional UIP and industrial production are imprecise; emerging-market breadth and the downcurve split are exploratory.

Specification selection remains the main uncertainty. The tenor, two-curve threshold, and release rule were assessed on the same return history. A deterministic real-time algorithm can overfit without fitted coefficients. Neighboring-rule grids show whether the baseline is isolated, but cannot exhaust discretion over tenors, cutoffs, countries, freshness, persistence, portfolio size, and timing.

Fifteen episodes limit independent common-state variation. Money-market accounting is not an executable forward strategy after basis, spreads, and timing costs. Predetermined equity beta discriminates among patterns but does not instrument for global growth. U.S. ACM components cannot decompose foreign term premia. Month-end dating prevents mechanical look-ahead but does not audit asynchronous closes, stale quotes, or historical vintages.

The result is a large in-sample predictive association ordered by rates and predetermined equity exposure. It does not establish causality or deployable performance. A frozen sample and feasible forward returns provide the next tests.

7 Conclusion

Synchronized fresh inversions across non-U.S. G10 yield curves predict a distinct currency-return state. Annualized carry-trade spot returns are 12.9 percentage points lower when the lagged state is active, and total carry returns fall from 6.5 to -4.9 percent. The known interest differential remains positive; exchange-rate losses overwhelm it. Median, quantile, and trimmed estimates show that the result is broader than a few crash months.

The cross-section supplies the economic interpretation. High-rate currencies lose relative to low-rate funding currencies, and bilateral losses become larger as predetermined equity-market beta rises. Persistent portfolio membership, a 0.72 alignment between rate and beta rankings, and a beta-sorted portfolio loss show that the state reprices a standing currency characteristic rather than merely rotating the carry portfolio. Directionally similar emerging-market responses extend the pattern beyond currencies used to construct the state, although they are not an independent replication.

This ordering supports the interpretation that synchronized inversions contain information about a shared international risk rather than only separate national disturbances. The evidence does not identify whether that information concerns global growth, coordinated policy, inflation, term premia, intermediary capacity, or a mixture. Slowly adjusting compensation is one coherent explanation, not a recovered structural mechanism.

The estimates use fifteen historical episodes and have not been evaluated in a frozen out-of-sample window. A prospective test should fix the state algorithm and data-vintage protocol before the next synchronized episode occurs.

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